

Lab Activity: Implementing a Secure OpenVPN Tunnel

Interface Check: Run ip addr. You should see a new tun0 interface with an IP in the 10.8.0.x range.

Routing Check: Run ip route. You should see a route for 192.168.1.0/24 pointing toward the tun0 interface.



[] Screenshot of systemctl status openvpn-server@server showing "Active (running)".

[] Screenshot of a successful ping 192.168.56.100 from the remote client.



Step 2: Critical Thinking & Performance Analysis

1. UDP vs. TCP: Why is UDP preferred for VPNs?

- **Answer:** UDP is preferred for VPNs to avoid "TCP Meltdown," a performance degradation issue that occurs when overlapping TCP retransmission timers conflict. TCP's built-in error correction and congestion control mechanisms become redundant and harmful when wrapped inside another TCP tunnel, whereas UDP allows the encapsulated application to manage its own reliability without this overhead.

2. PKI Security: If an employee leaves the company, we have their certificate, but they still have a copy on their laptop. How does a Certificate Revocation List (CRL) solve this?

- **Answer:** A Certificate Revocation List (CRL) acts as a server-side blacklist that invalidates the specific serial number of a certificate immediately upon revocation. This ensures that even if an ex-employee still possesses a physical copy of their certificate, the VPN server will reject any connection attempts made with that certificate.

3. Visibility: When the client pings 192.168.1.100, does the research server see the client's real IP or the VPN IP (10.8.0.6)? Why?

- **Answer:** The research server sees the VPN IP (10.8.0.6) rather than the client's physical IP address. This occurs because the VPN gateway performs network layer routing and address translation, effectively masking the client's physical identity behind the tunnel's virtual IP address.

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